

Absolute R² and Marginal Contribution of Multi-source Features

LightGBM

Random Forest

Spatial CV

Temporal CV

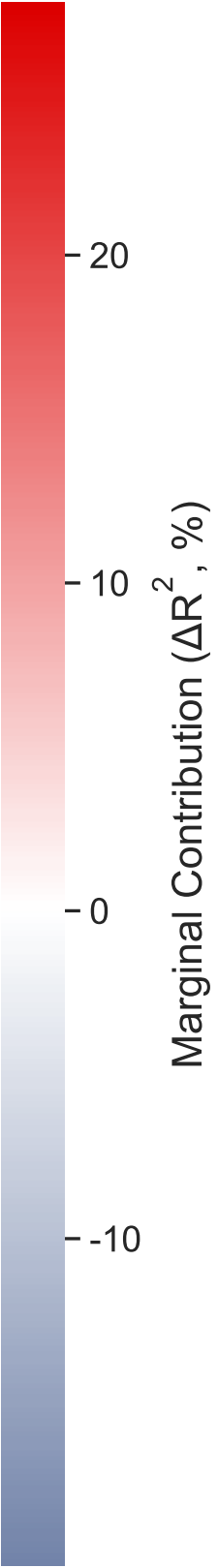
Spatial CV

Temporal CV

Feature Domains

Baseline
Buffer
Aggregated
Emission
Temporal
Topology
UCP
Land Use
POI

Spatial CV				Temporal CV				Spatial CV				Temporal CV				
Baseline	0.587 (Ref.)	0.622 (Ref.)	0.628 (Ref.)	0.606 (Ref.)	0.452 (Ref.)	0.556 (Ref.)	0.601 (Ref.)	0.486 (Ref.)	0.584 (Ref.)	0.629 (Ref.)	0.613 (Ref.)	0.590 (Ref.)	0.355 (Ref.)	0.485 (Ref.)	0.525 (Ref.)	0.497 (Ref.)
Buffer	0.634 (+4.7)	0.663 (+4.1)	0.709 (+8.1)	0.638 (+3.2)	0.498 (+4.6***)	0.604 (+4.7***)	0.694 (+9.3)	0.763 (+27.7***)	0.647 (+6.3)	0.708 (+7.9)	0.706 (+9.3*)	0.701 (+11.1)	0.474 (+12.0***)	0.573 (+8.8***)	0.661 (+13.6***)	0.719 (+22.2***)
Aggregated	0.574 (-1.4)	0.602 (-2.1)	0.673 (+4.4)	0.635 (+2.9)	0.508 (+5.6)	0.594 (+3.8***)	0.688 (+8.7*)	0.762 (+27.6***)	0.621 (+3.6)	0.672 (+4.3)	0.709 (+9.6*)	0.710 (+12.0)	0.457 (+10.2***)	0.542 (+5.6***)	0.644 (+11.9***)	0.703 (+20.6***)
Emission	0.624 (+3.7)	0.695 (+7.3)	0.720 (+9.2)	0.665 (+5.9)	0.454 (+0.2***)	0.627 (+7.1***)	0.672 (+7.0***)	0.585 (+9.9***)	0.670 (+8.6*)	0.732 (+10.3*)	0.737 (+12.4*)	0.715 (+12.5*)	0.434 (+7.9***)	0.587 (+10.2***)	0.634 (+10.9***)	0.610 (+11.3***)
Temporal	0.536 (-5.1)	0.619 (-0.3)	0.628 (-0.0)	0.601 (-0.5)	0.475 (+2.3***)	0.566 (+1.0***)	0.639 (+3.7)	0.607 (+12.1)	0.611 (+2.7*)	0.630 (+0.1)	0.602 (-1.1)	0.576 (-1.4)	0.427 (+7.3***)	0.475 (-1.1)	0.533 (+0.8)	0.523 (+2.6)
Topology	0.568 (-1.9)	0.599 (-2.3)	0.587 (-4.2)	0.406 (-20.0)	0.451 (-0.2***)	0.603 (+4.6***)	0.653 (+5.1*)	0.597 (+11.1***)	0.622 (+3.8)	0.631 (+0.2)	0.638 (+2.5)	0.595 (+0.5)	0.426 (+7.2***)	0.573 (+8.7***)	0.614 (+9.0***)	0.584 (+8.7***)
UCP	0.549 (-3.9)	0.569 (-5.3)	0.579 (-4.9)	0.555 (-5.1)	0.443 (-0.9***)	0.606 (+4.9**)	0.620 (+1.9*)	0.561 (+7.5***)	0.568 (-1.6)	0.605 (-2.4)	0.585 (-2.8)	0.565 (-2.5)	0.432 (+7.7***)	0.579 (+9.3***)	0.621 (+9.7***)	0.594 (+9.7***)
Land Use	0.523 (-6.4)	0.589 (-3.3)	0.488 (-14.0)	0.423 (-18.3)	0.467 (+1.5***)	0.622 (+6.6***)	0.620 (+1.9)	0.558 (+7.2***)	0.540 (-4.4)	0.601 (-2.8)	0.608 (-0.5)	0.576 (-1.4)	0.424 (+7.0***)	0.576 (+9.0***)	0.621 (+9.7***)	0.593 (+9.6***)
POI	0.523 (-6.4)	0.544 (-7.8)	0.519 (-10.9)	0.466 (-14.0)	0.440 (-1.3)	0.597 (+4.1***)	0.624 (+2.3)	0.581 (+9.4***)	0.600 (+1.6)	0.649 (+2.1)	0.622 (+0.9)	0.527 (-6.3)	0.389 (+3.4***)	0.528 (+4.3***)	0.569 (+4.4***)	0.539 (+4.1**)
	1h	24h	7d	31d	1h	24h	7d	31d	1h	24h	7d	31d	1h	24h	7d	31d



* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (One-tailed paired t -test across 58 monitoring stations)